

Python Programming Assessment

TOPIC: TERNARY & MATCH-CASE (GROUP 1)

Instructions:

1. This assessment contains 20 Multiple Choice Questions covering conditional expressions (ternary operator), structural pattern matching (match-case), and conditional pattern filtering (guards).
2. Read each question carefully and select the single best option.
3. No answers are provided in this question booklet. Write down your options on a separate answer sheet.

Q1. Which of the following is the correct order of a Python ternary operator?

- A) condition if true_value else false_value
- B) true_value if condition else false_value
- C) true_value else false_value if condition
- D) if condition else false_value if true_value

Q2. What is the output of the following code?

```
x = 10
```

```
result = "Even" if x % 2 == 0 else "Odd"
```

```
print(result)
```

- A) Odd
- B) Even
- C) True
- D) Error

Q3. What is the standard ternary operator (?) from languages like C++ or Java called in Python?

- A) Conditional Expression
- B) Switch Statement
- C) Match Statement
- D) Guard Expression

Q4. What keyword is used to start a structural pattern matching block in Python?

- A) switch
- B) select
- C) match
- D) case

Q5. Inside a match block, which keyword is used to define individual patterns to check against?

- A) check
- B) case
- C) item
- D) if

Q6. What is the purpose of the underscore (`_`) in a `case _:` statement?

- A) It matches only negative numbers.
- B) It acts as a catch-all wildcard (default case) if nothing else matches.
- C) It causes the program to skip the match block entirely.
- D) It represents an empty variable.

Q7. What will be the output of this simple match-case code?

```
color = "Red"
match color:
    case "Blue":
        print("Sky")
    case "Red":
        print("Fire")
```

- A) Sky
- B) Fire
- C) Error
- D) No Output

Q8. What keyword is used to add an extra condition (a guard) to a specific case?

- A) guard
- B) when
- C) if
- D) switch

Q9. In the statement `case x if x > 0:`, what is the part `if x > 0` called?

- A) Ternary operator
- B) Wildcard pattern
- C) Guard condition
- D) Unpacking sequence

Q10. What is the output of the following code?

```
age = 15
status = "Minor" if age < 18 else "Adult"
print(status)
```

- A) Adult
- B) Minor
- C) 15
- D) SyntaxError

Q11. Which operator is used within a single case line to mean OR?

- A) `&`

- B) |
- C) and
- D) or

Q12. What will be the output of the following code?

```
grade = "B"
match grade:
    case "A" | "B":
        print("Good Job")
    case _:
        print("Try Harder")
```

- A) Good Job
- B) Try Harder
- C) Error
- D) No Output

Q13. In a ternary operator expression, when is the else value chosen?

- A) When the condition is True.
- B) When the condition is False.
- C) When there is a syntax error.
- D) Always.

Q14. What will happen if a match-case statement finds NO matching patterns and there is NO case `_:` included?

- A) The program crashes with an error.
- B) The code simply moves on to the lines below the match block.
- C) It executes the very first case block anyway.
- D) It restarts the script.

Q15. What will be printed by this code snippet?

```
num = 5
match num:
    case x if x % 2 == 0:
        print("Even Number")
    case x if x % 2 != 0:
        print("Odd Number")
```

- A) Even Number
- B) Odd Number
- C) 5
- D) Error

Q16. Does Python's match-case automatically fall through to execute the next case block after finding a successful match?

- A) Yes, it executes all remaining cases.
- B) No, it executes exactly one matching block and stops looking.
- C) Yes, but only if you use the break keyword.
- D) It only falls through for numbers.

Q17. What will be the output of this code?

```
value = [10, 20]
```

```
match value:
```

```
    case [x, y]:
```

```
        print(x + y)
```

```
    case _:
```

```
        print("Not a pair")
```

- A) 10
- B) 20
- C) 30
- D) Not a pair

Q18. A guard condition inside a case block is evaluated:

- A) Before the structural pattern is checked.
- B) Only after the structural pattern matches successfully.
- C) Instead of checking the pattern.
- D) Only if the wildcard _ is used.

Q19. What will be printed by the following ternary expression?

```
is_logged_in = False
```

```
msg = "Welcome" if is_logged_in else "Please Login"
```

```
print(msg)
```

- A) Welcome
- B) Please Login
- C) False
- D) Msg

Q20. What is the output of this code with a guard constraint?

```
score = 45
```

```
match score:
```

```
    case s if s >= 50:
```

```
        print("Passed")
```

```
    case _:
```

```
        print("Failed")
```

- A) Passed
- B) Failed
- C) 45
- D) SyntaxError